

1) What does the article offer to your understanding of business ?

The article discusses Kevin Czinger's vision for a radical transformation in the manufacturing sector through Divergent 3D, a startup that utilizes advanced digital technologies like 3-D metal printing to miniaturize auto plants. Unlike traditional megafactories, Czinger envisions a network of small-scale urban factories producing low-cost, low-carbon vehicles in customizable batches, potentially bringing jobs back to communities. Instead of producing cars themselves, Divergent 3D focuses on licensing its innovative 3D printing technology to other manufacturers.

I believe Divergent 3D's business strategy embodies focused differentiation. This strategic approach involves targeting a niche group of consumers with distinctive offerings, which aligns with the company's innovativeness. In this case the company's innovation is the use of 3D printing technology in car manufacturing. I believe this to be focused differentiation as it focuses on the niche market of manufacturers providing unique service to gain competitive advantage. Specifically, I speculate that the target manufacturing companies could be Tesla, Toyota, Honda. I believe these companies would be a right fit as they already have hybrid electric cars which shows that they are adapting quickly with the market and matches the idea of environmentalism. Additionally, the unique way of penetrating the market with the business model of licensing its innovative 3D printing technology to other manufacturers is smart. As this idea of 3D printers in cars is new, this is a niche market providing unique service or product. Therefore, divergent 3D follows the strategy of focused differentiation.

2) What are the strengths AND weaknesses of the article ?

I believe Divergent 3D boasts remarkable strengths that set it apart in the automotive industry:

1. **Innovative Technology:** Divergent 3D's utilization of 3D printing technology for car manufacturing is phenomenal, presenting a unique and potentially disruptive approach to the industry. This innovative idea is undoubtedly a significant advantage. I see the potential in the company not immediately but in the long run.
2. **Cost Efficiency:** The company's commitment to cost efficiency is evident in its manufacturing process, promising lower production costs compared to traditional methods. As they highlighted, this cost reduction could make vehicles up to \$6,700 cheaper to build on average, potentially driving business growth. I strongly agree this will be a factor to gain competitive advantage in the market.
3. **Environmental Sustainability:** Divergent 3D's focus on reducing material waste and carbon emissions aligns perfectly with growing environmental concerns, appealing strongly to eco-conscious consumers. This would also be another factor that can differentiate to help the company.

However, it's essential to acknowledge the weaknesses:

1. **Market Acceptance:** Despite the groundbreaking technology, the concept of 3D-printed cars may face skepticism or resistance from consumers accustomed to traditional manufacturing methods. I don't think people can trust in machine-made cars at least in the next two decades, potentially hindering market adoption in the foreseeable future. This could pose a risk for manufacturers considering adopting Divergent 3D's technology, potentially impacting their existing sales due to concerns about consumer trust in machine-made cars.

2. **Regulatory Challenges:** I can see the compliance with automotive safety and quality regulations present as significant hurdles. I personally think the use of high-strength adhesive instead of welding raises concerns about safety.

3) What kinds of strategy- making do you see in this article ? This should be a professional analysis that reflects your understanding of the topics in this class.

As a strong advocate for environmentally friendly products, I believe that prioritizing Corporate Social Responsibility (CSR) and implementing Environmental, Social, and Governance (ESG) practices to be the primary strategy. Divergent 3D's manufacturing process, aimed at reducing material waste and carbon emissions, reflects this commitment to environmental sustainability. Additionally, their initiative to revitalize communities by creating job opportunities demonstrates social responsibility, although I remain cautious as traditional manufacturing methods typically employ more individuals than advanced, machine-based processes.

Another strategy worth noting is globalization, as emphasized by Czinger's ambition to expand globally. This vision underscores the company's aspirations for worldwide impact and growth. As there are car manufacturers around the world, I believe this would be a smart business strategy.

Furthermore, Divergent 3D's strategic partnerships with organizations like Groupe PSA, Horizon Ventures, and Altran Technologies indicate a collaborative approach aimed at driving innovation and fostering growth. I like this idea because it resembles Mergers and Acquisitions (M&A) in terms of the benefits it brings, such as support and relationship-building, yet it avoids the potential drawbacks associated with traditional M&A.